Discussion

The principal finding of this study is that the cortical transcriptome of feline chronic kidney disease reflects a shift in the tissue breadth of regulatory programmes. Genes that lose expression as disease advances are disproportionately those whose enhancers are active in the kidney and in no other profiled organ, while genes that gain expression are disproportionately those regulated by elements shared across organs. The association is substantial in both directions, survives adjustment for the number and signal intensity of enhancers per gene, and is measured in chromatin from healthy animals without reference to the expression data in which the gene classes were defined. It therefore constitutes evidence from a modality independent of the one that generated the question.

The most economical interpretation of that shift is cellular. Chronic kidney disease replaces tubular epithelium with immune infiltrate and fibrotic stroma, and the genes that define renal identity are precisely those regulated by kidney-restricted elements, while genes expressed by infiltrating leukocytes and fibroblasts are expressed in many tissues and regulated accordingly. Three analyses in this study converge on that reading. Coexpression modules were not reproducible under resampling except for a single axis that tracks proximal tubule marker expression at rho = 0.891 and accounts for 78% of the variance of two thirds of the repressed programme. Motif enrichment, tested against a background matched for expression and regulatory activity, was close to null. And the tissue-breadth association reproduces the same statement in chromatin. One alternative source of the composition signal can be excluded: although the cohort spans 2 to 20 years of age, none of the three marker scores varied with age, either marginally or conditional on disease group, so the loss of epithelial content is not an ageing effect within this sample. What the study does not establish is that the shift is only compositional, because bulk tissue cannot separate a change in the proportion of a cell type from a change in transcription within that cell type. The two hypotheses make the same prediction for every measurement available here.

This position is stronger than, but consistent with, the two previous analyses of the same dataset. The source study described metabolic reprogramming in feline CKD, reporting downregulation of fatty acid transport and oxidation, altered glucose and glutamine metabolism and upregulated hypoxia signalling. A subsequent secondary analysis of the cortical data reported that iron-handling, lipid-oxygenation and hypoxia-response transcripts moved in opposing directions rather than as a coordinated pathway, concluded that the data did not support inference of cell-specific mechanism, and noted that composition shifts might contribute materially. Neither study measured composition. The present analysis quantifies it, and in doing so offers an explanation for the incoherence the second study observed: gene sets defined by functional annotation cut across cell types, so in a tissue whose cellular proportions are changing they will contain genes moving in opposite directions for reasons unrelated to pathway regulation.

The observation concerning oestrogen-related receptors illustrates how narrow the inferential room is. Promoters of repressed cortical genes were enriched for ERR motifs, the ERRA and ERRB transcripts were themselves reduced, and the ERR-positive subset of repressed genes was specifically enriched for respiratory chain and mitochondrial inner membrane terms against a background of all repressed genes, while equivalent partitions on CTCF and SP1 motifs returned nothing. That configuration is what a genuine upstream regulator of the bioenergetic programme would produce. It is also what constitutive promoter architecture produces, since ERR elements occupy mitochondrial gene promoters irrespective of disease, and the stratified analysis showed the ERR–mitochondria association to be present with comparable strength among genes that did not change. Conditioning on mitochondrial annotation leaves a modest residual association between ERR-site presence and repression, at an odds ratio of 1.42. We report this as a constrained observation rather than a finding, because one named alternative remains untested: ERRA is highly expressed in proximal tubule, and the residual association could reflect tubular identity rather than regulatory activity. Mitochondrial annotation does not condition on tubular identity, and no external measure of tubular content was available for these animals.

Independently of the disease analysis, this study contributes the first application of the feline kidney regulatory atlas to a disease transcriptome, together with a gene-level annotation of promoter and enhancer architecture for the species. The atlas was deposited to support interpretation of non-coding sequence variants and has not been described in a publication of its own; its only previous reuse of which we are aware characterised epigenetic signatures of retroviral integrants [REF LaMere]. Applying it to disease expression is therefore a distinct use, and one that the resource supports without modification. The accompanying description of tissue breadth, in which 64% of promoter marks but 27% of enhancer marks are present in six or seven organs, reproduces in the cat a pattern established in better-characterised mammals and provides an internal validation of the classification.

One practical result deserves emphasis because it is a trap rather than a finding. The chromatin and expression datasets are aligned to different assemblies of the same genome, and those assemblies share chromosome names. Harmonising them by chromosome name retains 99.96% of intervals and produces no error, while yielding overlap statistics at the level of coincidence. We detected this only through a permutation control comparing observed promoter overlap against shuffled intervals, which returned 1.55-fold where the corrected annotation returns 8.59-fold. Any study combining these two resources, and more generally any study combining feline datasets deposited before and after 2021, should verify coordinate compatibility empirically rather than by inspecting identifier retention.

Several limitations constrain what these results support. The expression data derive from 21 cortical and 18 medullary samples collected at a single facility, and all inference about disease rests on that cohort. The chromatin data derive from two healthy adult male cats and describe the regulatory landscape of the normal cortex; no chromatin was profiled in diseased tissue, so no statement about remodelling of regulatory elements is possible, and the annotation describes the baseline architecture of loci that change rather than the state of those loci during disease. The chromatin and expression datasets come from entirely different animals. Peak sets were not filtered against a blacklist because none exists for this species, and we substituted a width-based filter. The kidney has the largest H3K27ac peak set among the seven profiled tissues, exceeding the smallest by a factor of 2.3, so the absolute proportion of kidney-specific enhancers is an upper bound, although this inflation is uniform across gene classes and cannot generate the differential association. Enhancer-to-gene assignment is topological, based on proximity within CTCF-delimited intervals, and was not validated by chromatin conformation data. Motif analysis used human position weight matrices and functional enrichment used human orthologs, since neither resource exists for the cat. The cohort spans a wide age range and includes animals of two and three years in the disease groups, which indicates aetiological heterogeneity within a sample otherwise characteristic of age-associated disease; the clinical entity captured here may therefore not be uniform. Adding age to the differential expression model leaves the cortical analysis unchanged but nearly doubles medullary detection while preserving effect estimates, so medullary counts reported here are conservative. Finally, the study was not conducted under a prespecified analysis plan, and several decisions were made in response to intermediate results; these are enumerated in the Methods.

Resolving the ambiguity that remains requires data of a kind not yet available for this species. Single-nucleus RNA sequencing of feline kidney across disease stages would separate change in cell proportion from change in transcription within cell type, and is the measurement the previous analysis of this dataset recommended without being able to perform. Per-animal histopathological scoring of glomerulosclerosis, interstitial fibrosis, inflammation and tubular atrophy, which was performed in the source study but reported only as group summaries, would provide an external measure of composition and permit adjustment that does not depend on the expression matrix itself. Chromatin profiling of diseased feline kidney would convert the annotation used here from a static reference into a comparison. Until such data exist, transcriptomic studies of bulk feline kidney should be read as descriptions of tissue composition as much as of cellular regulation.